

CMPG 324 - Operating Systems

Class Test 1

Examiner: Mr. JH Grobler

Date: 2024-07-26

Marks: 25

Time: 1 Hour

Name and Surname: _____

Student Number: _____

Signature: _____

Question 1 (6 Punte/Marks)

Afrikaans

Hoe verskil die bedryfstelsel wat op 'n bediener voorkom van die wat gebruik word op 'n skootrekenaar? Vergelyk die bedryfstelsels deur die tabelle hieronder te voltooi.

English

How does the operating system found on a server differ from one used on a laptop? Compare the operating systems by completing the tables below.

Eienskap/Attribute	Bediener/Server	Skootrekenaar/Laptop
Any software attribute, e.g. multiprogramming, multi-processing, multi-threading, 3rd party app support, etc. Must be comparable attribute within the context of both operating system types Hardware attributes, such as size of memory, battery, processing power, etc. do not qualify	1 mark if correct for attribute in the context of a server OS	1 mark if correct for attribute in the context of a personal computer OS
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Table 1: Question 1

Question 2 (3 Punte/Marks)

Afrikaans

Voltooi die volgende diagram deur die name van die procestoestande in te vul.

[3 marks] for correct process state names;

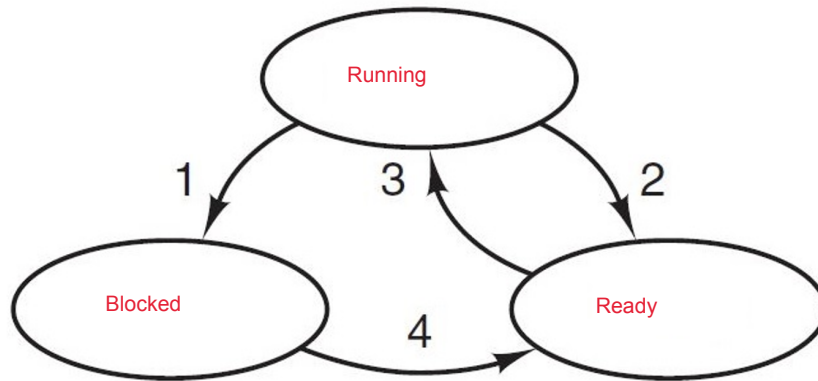


Figure 1: Question 2

English

Complete the following diagram by filling in the names of the process states.

Question 3 (4 Punte/Marks)

Afrikaans

Beskryf kortliks elkeen van die vier procestoestandoorgange wat in die diagram Figuur 1 getoon word. Maak seker dat jou beskrywings van toepassing is vir alle prosesse waar hierdie vier oorgange plaasvind.

English

Briefly describe each of the four process state transitions shown in Figure 1. Make sure your descriptions apply to all processes where these four transitions occur.

Oorgang/Transition	Beskrywing/Description
1	Process BLOCKS for I/O [1 mark]
2	Scheduler picks another process to run [1 mark]
3	Scheduler picks this process to run [1 mark]
4	I/O has completed and process is UNBLOCKED [1 mark]

Table 2: Caption

Question 4 (8 Punte/Marks)

Afrikaans

Vir elkeen van die volgende gevalle, toon aan watter proestoestandoorgang(e) die betrokke proses sal ervaar tydens die uitvoering van die betrokke kode. Maak gebruik van die nommers soos hulle in Figuur 1 voorkom. Identifiseer hoogstens die twee mees prominente oorgange per geval.

English

For each of the following cases, show which process state transition(s) the relevant process will experience during the execution of the relevant code. Use the numbers as they appear in Figure 1. Identify at most the two most prominent transitions per case.

Geval/Case	Nommer(s) van oorgang(e)/ Number(s) of transition(s)
Die volgende kode word op 'n tuisrekenaar uitgevoer / the following code is executed on a home computer: <pre> int sum = 0; for (int i = 0; i < 20000; i++) { int randomNum = (int) (Math.random() * 100); sum += randomNum; } </pre>	2, 3 [2 marks]
Die volgende kode word op 'n tuisrekenaar uitgevoer / the following code is executed on a home computer: <pre> File file = new File(fileName); Scanner scanner = new Scanner(file); int i = 0; while (scanner.hasNextLine()) { lines[i] = scanner.nextLine(); i++; } </pre>	1, 4 [2 marks]
'n Windows program met 'n grafiese koppelvlak voltooi sy take, en begin wag vir die gebruiker om nuwe instruksies via die grafiese koppelvlak te verskaf. / A Windows application with a graphic user interface completes its tasks, and starts waiting for the user to provide new instructions via the graphic user interface.	2, 3 [2 marks]
'n DMA beheerder gee te kenne dat al sy huidige I/O take vir die relevante proses voltooi is en die nodige stappe word geneem. / A DMA controller indicates that all its current I/O tasks have been completed for the relevant process and the necessary steps are taken.	4 [2 marks]

Table 3: Caption

Question 5 (4 Punte/Marks)

Afrikaans

Wat is vertakkings en hoe sal dit die procestoestande en procestoestandoorgange affekteer indien hulle gebruik word?

English

What are threads and how will they affect the process states and process state transitions if they are used?

Discuss what threading is [2 marks] and the impact it would have on the process states [2 marks],

e.g. process will not need to block as often, more efficient use of the CPU by the process, etc.

EINDE VAN KLASTOETS / END OF CLASS TEST